Aadarsha Gopala Reddy

+1 (740) 802-1776 | agr@agreddy.com | linkedin.com/in/agopalareddy | github.com/agopalareddy | agreddy.com

EDUCATION

Washington University in St. Louis — St. Louis, Missouri, USA
M.S. Computer Science — **August 2024 - May 2026**

- **Thesis:** Gopala Reddy, Aadarsha, "Toward Vehicle-Agnostic Driving Signatures for Cognitive Impairment Prediction from Naturalistic Driving Data" (2026). *McKelvey School of Engineering Graduate Student Theses & Dissertations.* 1341. https://doi.org/10.7936/g95n-fv35

Ohio Wesleyan University — Delaware, Ohio, USA
B.A. Computer Science & Data Analytics — **August 2020 - May 2023**

- **Minor:** Economics

RESEARCH EXPERIENCE

Washington University in St. Louis — St. Louis, Missouri, USA
Toward Vehicle-Agnostic Driving Signatures for Cognitive Impairment Prediction from Naturalistic Driving Data — **August 2025 - April 2026**
DRIVES Project (PI: Dr. Ganesh Babulal; Advisors: Dr. Alvitta Ottley, Dr. Nathan Jacobs)

- Built an end-to-end pipeline processing 26,968 participant-weeks of naturalistic driving data from 304 participants, deriving vehicle-agnostic weekly behavioral features and integrating demographic covariates for CDR status prediction.
- Compared six CDR prediction models (GRU-DANN, DANN, logistic regression, random forest, XGBoost, MLP) under leave-one-participant-out cross-validation; GRU-DANN achieved highest participant-level ROC AUC (0.599) and balanced accuracy (0.584).

Ohio Wesleyan University — Delaware, Ohio, USA
Opinion Survey on Artificial Intelligence in the Workplace — **January 2023 - May 2023**
Independent Study (Mentor: Dr. Nicholas Dietrich)

- Designed, fielded, and investigated a survey experiment on the impact of AI on 250 employees within each of four industries using survey data.
- Analyzed data using Tableau dashboards and R to identify trends and common opinions.
- Presented preliminary results on AI's positive impact on employee work at the OWU Spring Student Symposium.

Artificial Intelligence in Modern Board Games — Lost Cities AI — **May 2022 - July 2022**
Summer Science Research Program (Mentor: Dr. Sean McCulloch)

- Developed a digital version of the Lost Cities card game using Java.
- Implemented an intelligent agent that played the game against a human, winning 13 of 18 games in testing.

TEACHING & MENTORSHIP EXPERIENCE

Washington University in St. Louis — St. Louis, Missouri, USA
Graduate Teaching Assistant - CSE 5114: Data Manipulation and Management at Scale — **January 2026 - May 2026**

- Taught large-scale data pipeline concepts (Snowflake, Airflow, Spark, Kafka, Flink) to a graduate cohort, covering fault-tolerant design, streaming windowing strategies, and query optimization.
- Designed oral midterms as mock technical interviews, requiring students to reason through pipeline architecture decisions and troubleshoot system failures under real-world constraints.
- Mentored three capstone teams from raw open datasets to production-grade dashboards, providing weekly feedback that kept all teams on schedule through final presentations.
- Held weekly office hours to debug student pipeline implementations across Snowflake schema design, Airflow DAG failures, and Kafka consumer configuration.

HackWashU Databases Workshop — **September 2025**

- Designed and facilitated a hands-on databases workshop for beginner-to-intermediate developers, teaching through two complete projects rather than lectures.
- Guided participants through building a normalized SQLite database from the iTunes Search API and a full-stack Supabase Todo app with Row Level Security — covering relational databases, APIs, cloud Postgres, and access control.

Engineering Workshop by AGES & Science Coach — **September 2024**

- Delivered an introductory workshop on artificial intelligence to high school students, covering core AI concepts with hands-on demonstrations.

Ohio Wesleyan University Delaware, Ohio, USA
Computer Science Lab Assistant **January 2022 - May 2023**

- Tutored approximately ten students for over 6 hours per week, assisting with homework and exam preparation for introductory, intermediate, and advanced computer science coursework.
- Tested and graded assignments of around 35 students in introductory and intermediate computer science classes.

PRESENTATIONS

Ohio Wesleyan University Delaware, Ohio, USA
Spring Student Symposium **April 2023**
Artificial Intelligence Opinion Survey
Patricia Belt Conrades Summer Science Research Symposium **September 2022**
Artificial Intelligence in Modern Board Games

GRANTS, FELLOWSHIPS, & AWARDS

Bauer Leaders Academy (BLA) Leadership Development Badge, Washington University in St. Louis April 2026
Affiliated School Scholarship, Washington University in St. Louis May 2024
Dean's List, Ohio Wesleyan University May 2023
Mortar Board National College Senior Honor Society Membership, Ohio Wesleyan University April 2023
Golden Bishop Award - WCSA Best New Member, Ohio Wesleyan University April 2023
Dean's List, Ohio Wesleyan University May 2022
Florence Leas Sophomore Prize, Ohio Wesleyan University April 2022
First Place - 32nd Annual Spring Programming Contest, Denison University March 2022
International Baccalaureate Scholarship, Ohio Wesleyan University January 2020
Scholar, Next Genius Scholarship Foundation January 2020

PROFESSIONAL SERVICE & LEADERSHIP

Washington University in St. Louis St. Louis, Missouri, USA
Vice President of the Graduate-Professional Council (GPC) Chamber, Graduate and Professional Student Council (GPSC) **May 2025 - May 2026**

- Chaired the Constitution Committee (August 2025 - February 2026), coordinating stakeholder input to draft and ratify GPSC governing documents aligned with the Council's mission and university policies.
- Built committee membership rosters using ranked-choice preference allocation; represented the graduate student body to university leadership and supported the GPC President in advancing strategic council priorities.

Member, Graduate Student Affairs Advisory Board (GSAAB) **August 2025 - May 2026**

- Provided strategic feedback to the Vice Chancellor for Student Affairs and senior university leaders on critical issues impacting graduate student life and experience.
- Contributed to monthly discussions on student affairs policies and initiatives to enhance support systems and resources for the graduate student community.

Graduate Student Member, Center for Career Engagement (CCE) Student Advisory Board **August 2025 - May 2026**

- Guided the strategic direction of the Center for Career Engagement to ensure services align with diverse needs of graduate students from all academic backgrounds and career interests.
- Built community around career development initiatives and fostered collaboration between the CCE and graduate student body.

Treasurer, Umang (Indian Graduate Student Association) **June 2025 - October 2025**

- Managed club finances and organized cultural events and festivals, promoting cultural awareness and fostering community among Indian graduate students.

Treasurer, HackWashU **December 2024 - July 2025**

- Managed the club's finances, including budgeting and expense tracking; coordinated fundraising efforts and sponsorship outreach to support hackathons and coding events.

Ohio Wesleyan University Delaware, Ohio, USA
Budget Committee Senator, Wesleyan Council on Student Affairs (WCSA) **August 2022 - May 2023**

- Proposed and voted on senate bills while overseeing allocation of an annual budget exceeding $350,000 to campus individuals and organizations; received the Golden Bishop Award for WCSA Best New Member.

Vice President of Finance, Campus Programming Board **January 2022 - December 2022**

- Managed annual budgets exceeding $86,000 for campus-wide programming and secured $10,000 in additional annual funding through negotiation with WCSA.

PROJECTS

Washington University in St. Louis St. Louis, Missouri, USA

Red-Blue Visual Auto Defender: Automated Visual Jailbreak Generation and Explainable Defenses **August 2025 - December 2025**

- Built an automated red-blue teaming pipeline for vision-language model (VLM) security, addressing visual prompt injection (VPI) attacks against VLM-based agents.
- Generated visual jailbreak attack images by overlaying malicious instructions onto benign images and evaluated them against a target VLM (Gemma-3-4b-it) using semantic response analysis.
- Auto-generated deterministic, OCR-based Python defenses for each successful attack, validated with accuracy, precision, and recall metrics in an iterative refinement loop.

Multimodal Prediction of Alzheimer's Disease **August 2024 - December 2024**

- Developed a multimodal fusion approach for early detection of Alzheimer's Disease, combining MRI imaging and clinical/demographic data from the OASIS-1 dataset.
- Implemented CNNs using TensorFlow/Keras for brain MRI analysis alongside XGBoost and Random Forest models for tabular clinical data.
- Created a combined classifier integrating both modalities and documented the project in NeurIPS format with comprehensive performance evaluation metrics and visualizations.

Datacenter Cooling Optimization using Deep Reinforcement Learning **August 2024 - December 2024**

- Implemented multiple deep reinforcement learning algorithms (DDQN, PPO, SAC) integrated with EnergyPlus simulations to optimize datacenter cooling systems.
- Achieved 35.8% energy efficiency improvement over baseline using DDQN, addressing the underserved small to mid-sized datacenter market.
- Created a comprehensive framework integrating building energy simulation with reinforcement learning for dynamic thermal management.

MITxSureStart FutureMakers Create-a-Thon Program Remote

Parkinson's Disease AI Diagnosis Software **June 2022 - July 2022**

- Built a dual-input pipeline combining audio spectrograms (custom CNN for vocal tremor detection) and spiral-drawing images (Random Forest classifier with augmentation for micrographia detection).
- Implemented a weighted ensemble combining both modalities into a single Parkinson's diagnosis score.
- Designed a Flask web interface for user submissions of audio and drawing samples for real-time risk assessment.

OTHER PROFESSIONAL EXPERIENCE

Washington University in St. Louis St. Louis, Missouri, USA

Graduate Assistant, Taylor Family Center for Student Success **August 2024 - May 2026**

- Planned and ran three recurring engagement programs for first-generation, limited-income (FLI) students — Welcome Dinner, First-Gen Week, and End of Week Unwind — coordinating logistics, promotion, and on-the-day operations.
- Authored FERPA-compliant process manuals for engagement data collection and evaluation, giving center staff a standardized, repeatable framework for measuring FLI student participation.
- Analyzed semester survey responses and event attendance data, producing reports that directly informed the following cycle's programming decisions and outreach priorities.
- Managed the center's weekly newsletter; represented the department at off-campus excursions and FLI student trips.

Crittero, Inc. Remote
AI Engineer Intern **June 2025 - August 2025**

- Generated 100,000 labeled social media posts across seven user personas via time-series behavioral modeling in TypeScript, capturing post sequences, engagement timings, and content preferences.
- Trained a TensorFlow/Keras recommendation model achieving 80% accuracy (30-percentage-point gain over random baseline) with persona-type, content-category, and temporal embeddings, plus a collaborative-filtering cold-start fallback for new users.
- Built a configurable Python CLI test harness (700+ lines) with deterministic seeding and CSV export for reproducible model evaluation, using it to validate accuracy across manually tuned embedding-dimension and cold-start-threshold configurations.

Lab714 Boca Raton, Florida, USA
Data Analyst Intern **June 2023 - May 2024**

- Analyzed IoT environmental sensor data streams on AWS, building dashboards that surfaced chemical imbalances and equipment anomalies through automated threshold detection and trend analysis.
- Built AWS data ingestion pipelines to normalize and store continuous sensor telemetry from deployed devices, supporting real-time operational monitoring of deployed IoT fleets.
- Wrote SQL queries to aggregate historical sensor data by pool, date range, and chemical parameter, powering the trend analytics and anomaly detection shown in client dashboards.
- Designed and engineered the Figma-to-React UI for an all-in-one environmental sensor device, delivering a pixel-perfect, production-ready interface from initial wireframes.

TECHNICAL SKILLS & LANGUAGES

Programming Languages: Python, TypeScript, JavaScript, SQL, Java, C++, R, C#, Rust.

Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, React, Vue.js, Node.js, Express.js, FastAPI, Socket.IO, Streamlit.

Tools & Technologies: LaTeX, Git, Tableau, Power BI, AWS, Docker, PostgreSQL, MySQL, MongoDB, Snowflake, Apache Airflow, Spark, Kafka, Flink, Gemini/OpenAI APIs, Microsoft 365, Google Workspace.

Languages: Proficient in English, Kannada, Telugu; Conversational in Hindi.